

SATCOM Modernization Q&A

1. The typical weight limit in the DOD for a two-person lift is 112lbs per case. Is this weight limit applicable to this terminal?
 - a. Yes.
2. Is the contractor expected to provide the dual chain modem assemblies or are they provided? If the contractor is expected to provide them, can you please confirm the make and model?
 - a. Contractor/vendor will provide dual-chain modem assemblies that must be compatible with existing iDirect infrastructure.
3. Can you please provide the length of the Inter-facility Link cable (IFL Cable) needed?
 - a. The current cable is 1000 feet.
4. The SOO allows for a 1.2-2.4 meter antenna but also calls for a C Band capability. Is the intention to only operate on the SAT SIM or is the government accepting the fact that you will be training students to use C Band on a 1.2 meter, which would not be allowed on an actual satellite due to regulatory requirements?
 - a. The SOO calls for a 1.2-2.6m antenna to operate on SATSIM only.
5. Can you please provide the make/model of the existing, outdated, terminals in place? Are you looking for two different solutions or the same one for both? (For example, a Hawkeye 3 Lite in one and a Hawkeye 4 XY or AvL Technologies Model 2460 in the other, with each configurable to all required band frequencies?)
 - a. Current equipment is the AN/TSC-179 Ground Multi-Band Terminal (GMT). Terminals will be same type/model equipment, not two different model configurations.
6. Will the training terminals be used with a real satellite and beyond SATSIM purposes to enable actual on-orbit satellite training. If so, consider the next two questions.
 - a. Terminals will only be used on SATSIM equipment.
7. Please clarify SOO Sec 1.0 that states “This is a like-for-like replacement intended to maintain existing training capabilities and is not the acquisition of a new capability.” Given that the GMT systems are certified for operation on WGS satellites, can the Government confirm whether WGS certification will similarly be required for the proposed Dual-Hub terminals?

- a. No certification required.
8. Are the two contractor-provided laptops designed to operate the SATCOM terminal in a simultaneous dual-control configuration, or is the intention that one laptop controls only one or both aperture antennas at any given time?
- a. The requirement is for one laptop, per hub, per antenna. Each dual-hub terminal will be comprised of two independent antennas with independent LNB's.
9. Could you please clarify the meaning of 'operate on a low-power amplifier configuration'? Is the intent to source amplifiers with the minimum feasible output power, or amplifiers whose output can be sufficiently attenuated or muted to satisfy SATSIM operational constraints?
- a. Amplifiers must be capable of outputting between 1 watt and 10 watts to meet SATSIM constraints via either design, attenuation, or muting.
10. Antennas smaller than 1.8 meters are unlikely to satisfy the two-degree satellite spacing compliance criteria (e.g., per FCC Part 25 or ITU recommendations) for C-band. Is the terminal intended for SATSIM-only operation?
- a. Refer to question 6.
11. In Objective 4 you reference iDirect as an example. However, TDMA-based modems are generally not ideal for SATSIM operations. Did you mean a different modem manufacturer, say Comtech?
- a. Our current setup uses DMD-2050 modems (FDMA) and the iDirect modem (TDMA) using the SATSIM.
12. How far from the classroom and power source will the terminal be setup?
- a. The power source is co-located with the classrooms, which are approximately 100' from the terminals.
13. Please provide further clarity on what equipment is required, including modem(s). A block diagram would be helpful. (i.e., modem, NMS, HUB, and HUB Line Cards, Servers, Firewall, etc).
- a. Section 4 of the SOO specifies all required equipment, capabilities, and compatibilities.

14. Please clarify the verbiage “each terminal supporting two independent communication links (dual-hub) simultaneously.” With regard to:
- a. Dual terminal for multi orbit and the ability to orchestrate two simultaneous links
 - i. Not needed or required. One Dual-Hub terminal is comprised of two independent antennas with independent LNBs.
 - b. Two simultaneous links on one terminal (i.e., different pols or bands on WGS)
 - i. A single link on one terminal (perform a single link between the 2 hubs of one terminal).
15. As the requirement calls for the system to operate on a low power amplifier configuration for SATSIM operations, is this satisfied with attenuating WGS approved BUC, or does the OEM need to source a low power BUC?
- a. Refer to question 9.
16. Define dual-chain?
- a. The dual-use nature of satellites that can be adapted for various applications, including both civilian and military purposes.
17. Please clarify what level that schematics should be provided (i.e., LRU)?
- a. Refer to SOO section 5.
18. Does the terminal need a current WGS Certification?
- a. No certification required.
19. Would the customer value a modern design extensible to LEO and MEO operations? Would they consider a trade off for a modem, LEO capable antenna that might need to complete WGS certification?
- a. No.
20. For Ka-Band operation, does the terminal need to support full commercial and Mil bands or just Mil bands? Can the customer clarify minimum/desired frequency coverage per band?
- a. Our current equipment supports full commercial and Mil bands.
 - b. Our current capability is:
TX RF Freq:

C band: 5850 to 6425 MHz
X band: 7900 to 8400 MHz
Ku band: 13750 to 14500 MHz
Ka band: 30000 to 31000 MHz

RX RF Freq:

C band: 3400 to 4200 MHz
X band: 7250 to 7750 MHz
Ku band: 10950 to 12750 MHz
Ka band: 20200 to 21200 MHz

21. For C-Band operation, is circular polarization required or just linear?
 - a. C Band operations are not currently being applied, so just linear.
22. Noting the intended SATSIM operation, can the customer provide a minimum useful EIRP for live operations to drive BUC sizing?
 - a. Currently less than 1 watt but require adjustment up to 10 watts for future configurations
23. Where is the install location and what distance from the classroom? Will both terminals be installed at the same location?
 - a. Our current set up distance is 100 feet from the classroom. Both terminals are currently installed at the same location.
24. Please confirm the noted hub and modem components are Government Furnished Equipment?
 - a. No, the hubs and modems will be contractor provided.
25. Please clarify requirement for “dual-chain modem assemblies.”
 - a. The dual-chain modem assemblies are specialized communications devices designed to operate over multiple channels simultaneously for high-speed, secure and reliable communications. The systems we teach would be the operational NODES in field environment. See SOO for equipment requirements.
26. Please clarify the total number of antennas required for this configuration. Are two SATCOM terminals serving has HUB terminals that will support two REMOTE small aperture antennas, for a total of six (6) antennas? Or is the Government looking for two terminals, made up of two antennas each, for a total of four?

- a. The requirement is for two dual-hub terminals, each containing two antennas, for a total of four.
- 27. How many laptops does the Govt require?
 - a. The requirement is for four laptops, two per dual-hub terminal for a total of 4.
- 28. The SOO references operation on a "low-power amplifier configuration sufficient for SATSIM operations." Will the Government provide an existing satellite simulator, or is the contractor required to furnish SATSIM capability as part of the solution?
 - a. The LTT-03-W-198, Quad-Band SATSIM, is on-site.
- 29. What type of satellite network is the training curriculum designed to simulate (e.g., TDMA, SCPC, FDMA)? This is necessary to ensure the proposed modem configuration and hub compatibility align with the intended training scenarios.
 - a. Our current setup uses DMD-2050 modems (FDMA) and the iDirect modem (TDMA).
- 30. Does "dual-hub" operation require simultaneous transmission and reception across different frequency bands (e.g., C and Ku simultaneously), or does it refer to two independent channels operating within the same band at the same time?
 - a. Two independent channels operating within the same band at the same time. One Dual-Hub terminal is comprised of two independent antennas with independent LNBs.
- 31. The SOO references iDirect compatibility. Is the Government's intent to support both Comtech and iDirect modems - two of each per terminal - given that these platforms are not cross-compatible? Please confirm the required modem quantity and manufacturer mix per terminal.
 - a. The system configuration must be compatible with the existing iDirect hub infrastructure and must meet the capability requirements as specified within the SOO.
- 32. If iDirect modems are required, will the Government provide an existing iDirect hub, or is the contractor expected to provide hub infrastructure? Additionally, does the Government require iDirect hub support for all modulation types specified in the SOO (BPSK and QPSK minimum)?

- a. The contractor will utilize the Governments existing iDirect hub infrastructure. We do not require additional iDirect support, just compatibility as required in the SOO.

33. Is there a designated technical POC available to answer system-specific questions regarding the existing training infrastructure and curriculum integration requirements?

- a. A technical POC will be designated and available after award. Once the contractor/vendor provides the required “train-the-trainer” program, the government personnel will develop the in-house curriculum.

34. The SOO states that the contractor shall deliver two SATCOM terminals utilizing two small aperture antennas per terminal and operate across C, X, Ku, and Ka bands. Can the Government clarify if this means four antennas total (two per terminal) to support simultaneous operation across all four bands or will they be utilizing one band at a time across all antennas?

- a. Two antennas per terminal (each terminal requires two hubs for a total of four hubs and four antennas) all utilizing one band at a time across all antennas. One Dual-Hub terminal consists of two independent antennas with independent LNBs.

35. How will these terminals be used in the training environment, such as number of terminals and training sessions running simultaneously?

- a. Two dual hub terminals, up to two classes at the same time.

36. Will the modems be located on the antennas or in the classroom, and how far will the IFC need to run between equipment and classroom?

- a. Refer to question 12.

37. Can you extend the due date for proposal submissions?

- a. There is no interest in extending the proposal submission date at this time.